STA301 Statistics & Probability
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Solve By Vu Topper RM

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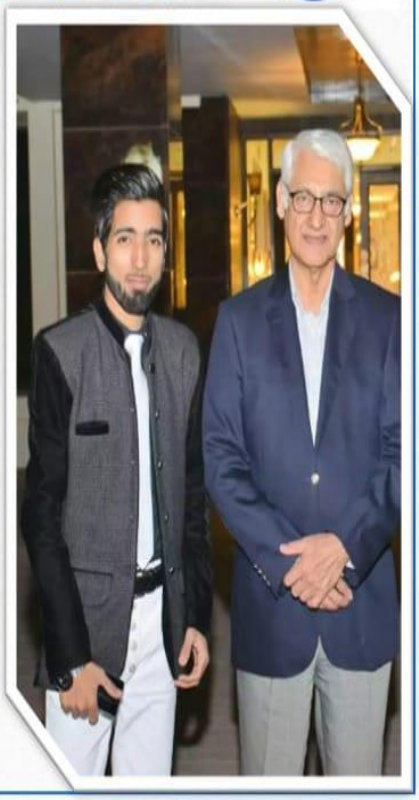
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Inferential statistics involves:

A. All of these
B. Estimation
C. Confidence interval
D. Testing of hypothesis

Conventionally, the probability of making a type-II error is denoted by:

A. ß (beta)
B. Ɵ (theta)
C. a (alpha)
D. б (sigma)

For a sample of size 20 taken from a normally distributed population with standard deviation equal to 5, a 90% confidence interval for the population mean would require the use of:

A. t = 1.328
B. t = 2.12
C. t = 1.729
D. z = 1.645

The level of significance is also called:

A. Acceptence region
B. Power of test
C. Type I error
D. Type II error

The proportion of working females in Pakistan is at most 0.40,the null hypothesis Ho<0.40 the alternative hypothesis H1 is:

A. P = 0.40
B. P > 0.40
C. P ≠ 0.40
D. P < 0.40

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Suppose that the workers of factory B believe that the average income of the workers of factory A exceeds their average income; the null hypothesis will be:

A. MeanA ≤ MeanB
B. MeanA < MeanB
C. MeanA ≥ MeanB
D. MeanA > MeanB

The proportion of cigarette smokers in Pakistan is greater than 20%, the null hypothesis Ho is:

A. P ≤ 0.20
B. P > 0.20
C. P < 0.20
D. P ≠ 0.20

The test statistic to test the U1 = U2 (U represent the mean of population)for normal population when n>30.

A. All of Above
B. T-Test
C. Z-Test
D. F-Test

The degrees of freedom associated with paired t-test ……

A. n1-2 and n2-2
B. n-1
C. n1+n2-4
D. n-2

As sample size goes up, what tends to happen to 95% confidence intervals?

A. They become more precise and narrow
B. They become more narrow

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C. They become wider
D. They become more precise

The confidence intervals become wide and less precise by
A. increasing the sample size
B. decreasing the sample size
C. decreasing the level of significance
D. None of these

For test of hypothesis Ho: μ1 = μ2 and H1: μ1 < μ2 ,the critical region at 0.05 level of significance and n > 30 is:
A. Z less and equal 1.96
B. Z> 1.96
C. Z>1.645
D. Z<-1.645

An industrial designer wants to determine the average amount of time it takes an adult to assemble an "easy to assemble" toy. A sample of 16 times yielded an average time of 19.92 minutes, with a population standard deviation equal to 5.73. Assuming normality of assembly times, what is a 95% confidence interval for the mean assembly time?
A. (14.1123, 22.7277)
B. (17.1123, 24.7277)
C. (17.1123, 22.7277)
D. (17.1123, 30.7277)

If we are testing H0: u 1 - u2 = 20 against H1: u 1 - u2 > 20 (exceeds 20).Then rejection region will be:
A. in Center
B. on both sides of Z=0
C. on right side of Z=0
D. on left side of Z=0

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The precision of an estimator can be increased by increasing the;

A. Size of sample
B. Size of population
C. Number of parameters
D. Number of estimators

Pairing is done only in case of …….observations.

A. Large
B. Small
C. Dependent
D. Independent

If we are testing H0 :P1 - P2 = 120 against H1 : P1 - P2 > 120 (exceeds 120).Then rejection region will be:

A. in Center
B. on left side of Z=0
C. on both sides of Z=0
D. on right side of Z=0

Which is the most appropriate statistical technique used to test for significant differences between two group means from data gathered by a repeated measures design?

Analysis of variance

For a sample p1= 0.4, n1 = 100 and for second sample p2= 0.6 and n2= 100, the pooled estimator pc is:

A. 0.60
B. 0.40
C. 0.55
D. 0.50

An alternative hypothesis is generally denoted by:

A. H3
B. H2

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C. H1
D. H0

Alpha is the probability of:
A. All
B. Type I error
C. Reject H0/H0 is true
D. Committing Type I error

If H0: P = 0.50 and H1: P > 0.50 then Critical region at 5% would be:
A. Z less than and equal to1.645
B. Z greater than 1.645
C. Z greater than and equal to 1.645
D. Z equal to 1.645

The median of the t-distribution is
A. Does not exist
B. Equals degrees of freedom
C. ONE
D. ZERO

If P (6.2< ? <12.2) =0.95 then
A. We are 5% confident that true value of the parameter fall in this interval
B. We are 95% confident that true value of the parameter fall in this interval
C. This is wrong statement
D. All are correct

For a particular hypothesis test, Alpha=0.09 and Beta=0.03, what is the value of Type I error?
A. 0.97
B. 0.03
C. 0.09

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D. 0.91

The t-test for independent samples assumes which of the following?
A. Groups have equal medians
B. Scores are dependent
C. Groups have equal means
D. Scores are independent

A deserving player was not selected in the cricket team. It is an example of:
A. Type-I error
B. Type-Il error
C. Type-Ill error
D. Type-IVerror

For a particular hypothesis test, Alpha=0.10 and Beta=0.15. What is the value of Type I error?
A. 0.85
B. 0.10
C. 0.90
D. 0.15

Suppose H0 : u 1 = u2 and H1:u1 >u 2 and Critical value is z= 1.645. If calculated value of Z = 1.32,then what will be your conclusion?
A. Reject Ho
B. Impossible to decide
C. None of the above
D. Accept Ho

The density of the t-distribution reaches its maximum at t=_____:
A. 3
B. 2
C. 0
D. 1

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t-distribution can be used to find the confidence interval for:

A. Population mean
B. Population variance
C. Populating proportion
D. Difference between population proportions

When sample size is to be considered large, population standard deviation can be replaced by

A. Sample Mean
B. Sample Variance
C. Population variance
D. Sample Standard Deviation

If the NULL hypothesis says that the average height of Pakistani soldiers exceeds the average height of American soldiers by not more than 3 inches. What will be its ALTERNATIVE hypothesis?

A. P > 3
B. none of these
C. P < 3
D. P = 3

To test the hypothesis about the difference of means for large samples, what test statistic will be used?

A. Chi-square
B. z
C. t
D. f

For a particular hypothesis test, Alpha=0.10 and Beta=0.15. What is the value of Type II error?

A. 0.85
B. 0.10
C. 0.90

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D. 0.15

In testing H0: $\mu = 100$ against H1: $\mu \leq 100$ at the 10% level of significance,H0 is rejected if:

A. 100 is contained in the 90% confidence interval
B. The p-value is greater than 0.10
C. The value of the test statistic is in the acceptance region
D. The p-value is less than 0.10

Which of the following statements is NOT true?

A. The t distribution is more spread out than the standard normal distribution
B. As the degrees of freedom get smaller, the t-distribution's dispersion gets smaller
C. The t distribution is symmetric about zero
D. The t distribution is mound-shaped

Alpha is the probability of ……..

A. Making Type I error
B. Accepting Ho when it is true
C. Rejecting H1 when it is wrong
D. Making Type II error

By increasing sample size, _____ can be decreased:

A. Both Alpha and Beta
B. Beta
C. Alpha
D. None of these

The location of the critical region depends upon:

A. Value of alpha
B. Alternative hypothesis
C. Null hypothesis

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D. Value of test-statistic

A _____ is a range of numbers inferred from the sample that has a certain probability of including the population parameter over the long run.

A. Lower limit
B. Hypothesis
C. confidence interval

For a sample p1 = 0.2, n1= 100 and for second sample p2= 0.5 and n2= 200, the pooled estimator pc is:

A. 0.35
B. 0.30
C. 0.40
D. 0.45

All other things remaining constant, a 99% confidence interval for a population mean μ will _____ the 95% confidence interval for μ.

A. be narrower than
B. None of the above
C. be narrower than
D. be wider than

Suppose H0: p =0 H1:p < 0 and Critical value is z = - 2.33. If calculate the value of Z = - 2.60,then what will be your conclusion?

A. Impossible to decide
B. None of the above
C. Accept Ho
D. Reject Ho

To test the hypothesis about the difference of means for small samples, test statistics will be used:

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A. T-test
B. F-test
C. Z-test
D. All of these

The t-distribution is applicable for testing about the population mean, when population standard deviation is:
A. Known
B. Unknown
C. Less than 40
D. Greater then 40

If we are testing H0 : P1 < P2(or) P1 - P2 < 0 against H1 : P1 > P2(or) P1 - P2 > 0 and the tabulated value of Z = 1.645 and Z calculated value = 2.63 then what will be the conclusion: \end{gathered} \]
A. Reject Ho
B. Impossible to decide
C. Accept Ho
D. None of the above

If we have the following information, P= 0.16, q = 0.84, Z=1.96 and e = 0.05, then sample size will be:
A. 205
B. 199
C. 200
D. 207

To determine a sample size in estimating population mean, we use the _____value.
A. X
B. F
C. T
D. Z

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If we are testing H0: P1=P2 against H1:P1 > P2(or P1 -P2 > 0 and tabulated value of z = 1.96 and Z calculated value = 1.63 then what will be the conclusion:

A. Reject Ho Reject H0
B. Accept Ho
C. Impossible to decide
D. None of the above

Which one provides the basis for hypothesis testing?

A. Null hypothesis
B. Critical value
C. Test-statistic
D. alternative hypoetheis

_____ are equivalent.

A. Type-II error and Level of confidence
B. Type-I error and Level of significance
C. Type-I error and Level of confidence
D. Type-II error and Level of significance

Which of the following is not a criterion for a good problem situation?

A. It must be clearly structured and defined.
B. It must be such that its solution will be benefited by group effort.
C. It must be appropriate to the students' level of development.
D. It must be authentic.

The term 1- β is called

A. Level of the test
B. Power of the test
C. Size of the test
D. Critical region

Critical region is also known as:

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A. None of these
B. Neither a and b
C. Rejection region
D. Acceptence Region

t-distribution is used to test the hypothesis about:
A. variance
B. Proportions
C. ratio of variance
D. Mean

We can apply t-distribution to test the hypothesis, when
A. Sample is large
B. Sample is small
C. Population variance is known
D. Population variance is unknown and Sample is small

The critical region is computed from the value of:
A. Beta
B. Alpha
C. 1-Beta
D. 1-alpha

A failing student is passed by an examiner. It is an example of:
A. Type I error
B. Type II error
C. Correct decision
D. No information regarding student exams

"A point estimate plus/minus a few times the standard error of that estimate". This statement represents:
A. Critical value
B. Critical region

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C. Acceptance region
D. Confidence interval

If we reject the true null hypothesis using t-distribution, we might be making:

A. Type I error
B. Type II error
C. Correct decision
D. No information regarding student exams

A random sample of n=25 values gives sample mean 83. Can this sample be regarded as drawn from a normal population with μ= 80 and s= 7? In this question the alternative hypothesis will be:

A. H1: μ = 80
B. H1: μ ≠ 80
C. H1: μ > 80
D. H1: μ <80

ß is the probability of:

A. Accept H0/H0 is false
B. Accept H0/H0 is true
C. Reject H0/H0 is false
D. Reject H0/H0 is true

For a particular hypothesis test, α=0.05and β=0.05. The power of test is equal to:

A. 0.05
B. 0.50
C. 0.90
D. 0.95

Mean of t-distribution will not exist when d.f(degree of freedom) will equal to:

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A. 4
B. 2
C. 3
D. 1

To test the hypothesis about the difference of means, which test we use for testing?
A. z
B. both a and b
C. f
D. t

Confidence interval is also known as
A. Level of Significance
B. None of these
C. Margin of error
D. Point estimator

Accept H0/H0 is false, is the _____.
A. Type I error
B. Type II error
C. Correct decision
D. None of the above

If we are testing H0 : u 1- u 2 = 10 against H1: u 1 - u 2 < 10.Then rejection region will be:
A. on right side of Z=0
B. on both sides of Z=0
C. on left side of Z=0
D. in Center

The Test-statistic is used to decide whether to_____a null-hypothesis.

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A. Accept or Reject
B. Non of these
C. Accept
D. Reject

Which test is preferred when you want to compare two population means where you have two samples in which observations in one sample can be paired with observations in the other sample?
A. Chi-square
B. ANOVA
C. T-test
D. Paired t-test

If a significance level of 1% is used rather than 5%, the null hypothesis is:
A. Just as likely to be rejected
B. More likely to be rejected
C. Less likely to be rejected
D. None of the these

For a particular hypothesis test, Alpha=0.05 and Beta=0.15. What is the level of significance?
A. 0.85
B. 0.15
C. 0.05
D. 0.95

Suppose H0: u 1= u2 and H1:u 1 < u 2 and Critical value is z= - 1.645. If calculated value of Z= - 2.08,then what will be your conclusion?
A. Accept Ho
B. None of the above
C. Reject Ho
D. Impossible to decide

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When examining group difference where no direction of the difference is specified, which of the following is used?

A. One-tailed test
B. Two-tailed test
C. Directional hypothesis
D. Difference of modes test

The end points that bound the confidence interval are called;

A. Lower limits
B. Upper limits
C. Bounded limits
D. Lower and upper limits

For a particular hypothesis test, Alpha=0.07 and Beta=0.05. What is the level of significance?

A. 0.95
B. 0.07
C. 0.05
D. 0.93

What is the rule of thumb for a sample size "n" to be considered large.

A. n<30
B. n>30 or n=30
C. n=30
D. None of the above

The probability of success changes from trial to trial, is the property of:

A. Poisson experiment
B. Binomial experiment
C. Hypergeometric experiment
D. Both binomial & hypergeometric experiment

The normal curve is:

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A. The typical frequency curve for psychological data.
B. A convenient fiction.
C. The standard against which all other distributions are assessed.
D. A bell-shaped distribution

Binomial distribution has two parameters; which are:
A. None
B. X and n
C. p and n
D. p and q

If in a lottery, one thousand tickets are sold. The chance of a ticket for wining the lottery is:
A. Mutually exclusive
B. Not equally likely
C. Equally likely
D. No information is given

A sales firm receives different calls from clients. If someone says, today it will receive "at least 3 calls". How many calls will be received?
A. 0, 1, 2, 3, 5, . . .
B. 0, 1, 2, 3
C. 3, 4, 5, . . .
D. 1, 2, 3

The number of trials in hypergeometric distribution is:
A. Small
B. Large
C. Not fixed
D. Fixed

It is sometimes possible to obtain approximate probabilities associated with values of a

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random variable by using the different probability distribution of a different random variable. For example, binomial probabilities use the Poisson probability function, binomial probabilities using the normal etc. In order for the Poisson to give "good" approximate values for binomial probabilities we must have the condition(s) that:

A. The population size is large relative to the sample size
B. The probability, p, is small and the sample size is large
C. The probability, p, is close to 0.5 and the sample size is large
D. The probability, p, is close to 0.5 and the population size is large

If the second moment ratio is less than 3 the distribution will be:

A. Leptokurtic
B. Mesokurtic
C. Platykurtic
D. None of these

Which of the following is a characteristic of a binomial probability experiment?

A. P(success) = P(failure)
B. Each trial has more than two possible outcomes
C. The binomial random variable x is the count of the number of trials that occur
D. The result of one trial does not affect the probability of success on any other trial

Which of the following is not a property of a binomial experiment?

A. All of the above
B. the experiment consists of a sequence of n identical trials
C. each outcome can be referred to as a success or a failure
D. the probabilities of the two outcomes can change from one trial to the next

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If a random variable X denotes the number of heads when we toss a fair coin 5 times, the X assumed the values:

A. 0,1,2,3
B. 1, 5, 5
C. 1, 2,3,4,5
D. 0, 1, 2,3,4,5

The lottery conducted in various countries for purposes of money-making provides a good example of:

A. Binomial distribution
B. Normal distribution
C. Poisson distribution
D. Uniform distribution

The number of typing errors per page in a book follows:

A. Normal Distribution
B. Poisson Process
C. Binomial Distribution
D. Hyper geometric Distribution

In a binomial experiment successive trials are:

A. Both
B. None
C. Dependent
D. Independent

In a binomial experiment with three trials, the variable can take:

A. 2 values
B. 4 values
C. 5 values
D. 3 values

If p is very small and n is considerably large then we shall apply the:

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A. Binomial distribution
B. Hypergeometric distribution
C. Poisson distribution
D. Exponential distribution

The location and shape of the normal curve is (are) determined by:

A. Mean
B. Variance
C. Mean & variance
D. Mean & standard deviation

The total area under the normal curve is:

A. 0
B. 1 **Page 186**
C. 0.5
D. 0.75

Poisson distribution can be used to approximate the hyper geometric distribution when; n <0.05N, n>20 and

A. p>0.05
B. p>0.10
C. p<0.05
D. p<0.10

The function F(x) gives the probability of the event that X takes a value a value.....

A. P(X≤x)
B. P(X>x)
C. P(X<x)
D. P(X=x)

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In hyper geometric distribution; experiment can be repeated a

A. Depends upon the situation
B. variable number of times
C. None of the above
D. fixed number of times

The hypergeometric probability distribution has parameters:

A. N, n, k
B. N, N-K, n
C. N, n
D. N, P, n

Mean of binomial distribution is:

A. n/p
B. np
C. n
D. p

In a binomial experiment ,probability of success changes on:

A. 10th trial
B. Each trial
C. 30th trial
D. Second trial

In a bivariate probability distribution of X and Y,If f(0,1)=3/14, g(0)=5/14 and h(1)=3/7,then

A. Both of these
B. None of these
C. X and Y are statistically independent
D. X and Y are not statistically independent

The values of moment ratios b1 and b2 of normal distribution are:

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A. 0 and 1
B. 0 and 2
C. 0 and 3
D. 0 and 4

If value of p is smaller or lesser than 0.5 then binomial distribution is classified as:

A. skewed to left
B. skewed to infinity
C. skewed to right
D. skewed to integers

E(X + Y) = ?

A. E(X + Y) = E(X) E(Y).
B. E(X + Y) = E(X) + E(Y).
C. E(X + Y) = E(X) + E(Y).
D. None

The mean of Uniform distribution defined on the interval [a,b] is:

A. (a−b)
B. a+b2
C. (a+b)
D. a−b2

In a normal distribution how much area lies between μ±σ

A. 80%
B. 65%
C. 68%
D. 75%

We can apply binomial distribution formula, when we are drawing a sample from finite population without replacement and the sample size

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"n" is not more than ……..% of the population size N.

A. 8%
B. 5%
C. 10%
D. 1%

For good approximation of Poisson distribution to the binomial distribution, which of the following condition (s) is/are required:

A. The population size is large relative to the sample size
B. The probability, p, is small and the sample size is large
C. The probability, p, is close to .5 and the sample size is large
D. The probability, p, is close to .5 and the population size is large

Poisson distribution is never:

A. Symmetrical
B. Asymmetrical
C. Negatively skewed
D. Positively Skewed

If X and Y are independent, then Cov (X, Y) is equal to:

A. 0
B. 1
C. None
D. 0 and 1

In a binomial experiment:

A. None of the above
B. Number of trials "n" is a fixed
C. Number of trials "n" is a variable
D. Number of trials "n" is not specified

The location parameter of normal distribution is:

A. Variance

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B. Mean
C. Range
D. Standard Deviation

If total probability of joint probability distribution f(x,y) is ONE then what will be the total probability of its marginal pdf h(y)?
A. One
B. Less than one
C. Depend on given values
D. Anywhere between zero and one

The hypergeometric random variable is a(an):
A. Undefined
B. Independent
C. Discrete variable
D. Continuous variable

The correlation coefficient:
A. makes the scatterplot redundant.
B. should be interpreted alongside the relevant scatterplot.
C. all of these.
D. can be expressed as a percentage.

Suppose 60% of a herd of cattle is infected with a particular disease. Let Y = the number of non-diseased cattle in a sample of size 5. the distribution of Y is:
A. Poisson with u = .6
B. Binomial with n = 5 and p = 0.4
C. Binomial with n = 5 and p = 0.5
D. Binomial with n = 5 and p = 0.6

Density function of normal distribution attains its maximum value when the value of random variable X is:

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A. X = Standard Deviation
B. X = Mean
C. All above
D. X = Variance

The mean of the continuous probability function f(x) can be obtained by
A. Multiplying the x by its function f(x)
B. Integrating the product xf(x)
C. Integrating the function f(x)
D. Integrating the x

In a bivarite probability distribution of X and Y,If E(X)=3 and E(Y)=5, then E(X-Y)= ?
A. -2
B. 15
C. None of these
D. 2

In case of bivariate distribution, co-variance of distribution is represented:
A. E(X.X)-E(X).E(Y)
B. E(Y.X)-E(X).E(Y)
C. E(Y.Y)-E(X).E(Y)
D. E(X^2)-[E(X)]^2

What will be the value of C.V if S.D = 0.5 and mean = 1.
A. 50%
B. 45%
C. 30%
D. 10%

Which of the following is the property of hypergeometric experiment?
A. Sampling is performed without replacement

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B. p remains constant from trial to trial
C. n is not fixed
D. Successive trials are independent

If total probability of joint probability distribution f(x,y) is ONE then what will be the total probability of its marginal pdf g(x)?

A. One
B. Half of the joint pdf
C. Anywhere between zero and one
D. Less than one

If the first moment ratio is equal to 0, then the distribution will be:

A. Positively Skewed
B. Symmetrical
C. Leptokurtic
D. Negatively Skewed

For the Poisson distribution P(X = 1) = e−2211! the mean value is:

A. 0
B. 2
C. 1
D. 3

The shape parameter of normal distribution is:

A. Standard Deviation
B. Median
C. Mean
D. Mode

In hyper geometric distribution probability of success always:

A. None of the above
B. sometimes varies and sometimes is constant
C. Varies from trial to trial

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D. is constant from trial to trial

The number of parameters in Poisson distribution is (are):
A. 0
B. 1
C. 2
D. 3

Using the normal approximation to the binomial distribution with n= 3 and p= 0.0571 the value of mean is:
0.5133
0.1713 **Page 235**
0.2132
0.1923

Which type of data is used in the binomial distribution?
A. Continues data only
B. Both discrete and continuous
C. Discrete data only
D. Qualitative data only

The relationship between the mean and variance of Poisson distribution is:
A. Mean >Variance
B. Mean = Variance
C. Mean <Variance
D. Mean = Standard deviation

In a normal distribution how much area lies between $\mu \pm 2\sigma$
A. 65%
B. 75%

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C. 95%.45%
D. 94%

A correlation coefficient:
A. all of these.
B. is a sort of index of how close the points of a scattergram deviate from the best fitting straight line through those points.
C. tells you the direction of the slope of the scattergram.
D. efficiently summarises some of the information in a scatterplot.

Question No:1 (Marks:1) Vu-Topper RM
Uniform distribution is a:
Mixed
Discrete distribution
Continuous Distribution
None of these

Question No:2 (Marks:1) Vu-Topper RM
A confidence interval has a specified probability_____ of containing the true value of the parameter.
A. α
B. 1- α
C. 1+ α
D. α/2

Question No:3 (Marks:1) Vu-Topper RM
When the event occur randomly over a specified interval of time, space or length; the distribution is known as:
A. Poisson Process Page 217
B. None of the above
C. Poisson Distribution
D. Hyper geometric distribution

Question No:4 (Marks:1) Vu-Topper RM

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Which of the following is true for the Poisson distribution:

A. mean > variance
B. mean < variance
C. mean = variance Page 233 ok
D. mean = standard deviation

Question No:5 (Marks:1) Vu-Topper RM

Poisson distribution is the limiting form of binomial distribution when n, the number of trials is large and p, the probability of success is

A. very Small Google
B. Very large
C. Moderately small
D. Moderately large

Question No:6 (Marks:1) Vu-Topper RM

The standard normal distribution has it mean and variance equal to:

A. 0 and 0 respectively
B. 1 and 1 respectively
C. 0 and 1 respectively Google ok
D. 1 and 0 respectively

Question No:7 (Marks:1) Vu-Topper RM

When an interval is constructed, what is the probability of occurrence of parameter in it?

A. 1
B. 0
C. 50
D. Either 1 or 0 Page 263

Question No:8 (Marks:1) Vu-Topper RM

When we draw the sample with replacement, the probability distribution to be used is:

A. Poisson

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B. Binomial
C. Hypergeometric
D. Binomial & hypergeometric

Question No:9 **(Marks:1)** **Vu-Topper RM**
For the Poisson distribution P(X = 1) = e−0.1350.13511! the mean value is:
A. 1
B. 0.135 **Page 222** **ok**
C. -0.135
D. e−0.135

Question No:10 **(Marks:1)** **Vu-Topper RM**
A quantity obtained by applying certain rule or formula is known as
A. Estimate **Page 264**
B. Estimator
C. Parameter
D. Proportion

Question No:11 **(Marks:1)** **Vu-Topper RM**
A random sample of n = 6 has the elements 6, 10,13,14,18 and 20. What is the point estimate of the population mean?
A. 11
B. 12
C. 11.5
D. 13.5

Question No:12 **(Marks:1)** **Vu-Topper RM**
An urn contains 4 red balls and 6 green balls. A sample of 4 balls is selected from the urn without replacement. It is the example of:
A. Binomial distribution
B. Hypergeometric distribution **Page 219**
C. Poisson distribution

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D. Exponential distribution

Question No:13 (Marks:1) Vu-Topper RM

A statistic whose standard error decreases with an increase in the sample size will be .....

A. None
B. Efficient
C. Unbiased
D. Consistent

Question No:14 (Marks:1) Vu-Topper RM

The larger the standard error, the ..... the confidence interval.

A. Wider
B. Accurate
C. Inaccurate
D. Narrower

Question No:15 (Marks:1) Vu-Topper RM

The mean deviation of the normal distribution is approximately:

A. 7/8 of the S.D
B. 4/5 of the S.D
C. 3/4 of the S.D ok
D. 1/2 of the S.D

Question No:16 (Marks:1) Vu-Topper RM

Mean and variance of a Poisson Distribution are:

A. Equal Google
B. Not equal
C. Mean is greater than variance
D. Variance is greater than mean

Question No:17 (Marks:1) Vu-Topper RM

If the first moment ratio is greater than 0, then the distribution will be:

Moment

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Ratios Page 222

Question No:18 (Marks:1) Vu-Topper RM

A statistic whose standard deviation decreases with an increase in sample size will be called:

A. None
B. Efficient
C. Unbiased
D. Consistent

Question No:19 (Marks:1) Vu-Topper RM

If we have two normally distributed populations then sampling distribution of the difference of sample means will be:

Normal distribution

Question No:20 (Marks:1) Vu-Topper RM

Sample Proportion is a /an_____ estimator of population mean.

A. Unbiased
B. Consistent
C. None of these
D. Unbiased and Consistent estimator

Question No:21 (Marks:1) Vu-Topper RM

The range of the binomial distribution is:

A. 1,2, …….n
B. 0, 1, 2, …… n
C. 0, 1, 2, …… x
D. 0, 1, 2, …… 100

Question No:22 (Marks:1) Vu-Topper RM

In Binomial distribution the sample size is considered to be sufficiently large, if both np and nq are greater than or equal to:

A. 3
B. 5 **Google**

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C. 10
D. 15

Question No:23 (Marks:1) Vu-Topper RM
The probability distribution of the proportion of successes in all possible samples is called the:

A. Sampling distribution Page 237
B. Sampling probability distribution
C. Sampling distribution of sample proportions
D. Sampling distribution of Population proportions

Question No:24 (Marks:1) Vu-Topper RM
Consider a large population with a mean of 160 and a standard deviation of 20. A random sample of size 64 is taken from this population. What is the standard deviation of the sample mean?

A. 1.654
B. 2.500
C. 3.125
D. 3.568

Question No:25 (Marks:1) Vu-Topper RM
providing as an estimate a single value calculated from the sample that is likely to be close in magnitude to the unknown parameter is known as

A. Estimation
B. Point Estimation Google
C. Interval Estimation
D. None of the above

Question No:26 (Marks:1) Vu-Topper RM
The width of the confidence interval is called_____ of the estimate.

A. Biased
B. Accuracy
C. Precision Google

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D. Unbiasedness

Question No:27 **(Marks:1)** **Vu-Topper RM**

To determine a sample size which test we use …….

A. F

B. Z

C. T

D. None of these

Question No:28 **(Marks:1)** **Vu-Topper RM**

The central limit theorem states that the mean of the sampling distribution of the means is equal to the population:

A. Mean

B. Random Error

C. Standard error

D. Standard deviation

Question No:29 **(Marks:1)** **Vu-Topper RM**

In a Geometric distribution, Maximum Likelihood Estimator (MLE) for proportion (P) is equal to:

A. Sample mean

B. Sample variance

C. Reciprocal of the mean

D. Reciprocal of the sample variance

Question No:30 **(Marks:1)** **Vu-Topper RM**

Sampling distributions are:

A. None oof the given

B. Refers to the margine of error

C. The probability distribution of sample statistic **Google**

D. The probability distribution that is assigned to population

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parameters.

Question No:31 **(Marks:1)** **Vu-Topper RM**

In binomial distribution, formula of calculating mean is:

A. U=np
B. U=pq
C. U=qn
D. U=p+q

Question No:32 **(Marks:1)** **Vu-Topper RM**

Ideally, the width of confidence interval should be

A. 0
B. 1
C. 98
D. 100

Question No:33 **(Marks:1)** **Vu-Topper RM**

The binomial distribution is negatively skewed when:

A. p>q **Page 215** **ok**
B. p=q
C. p<q
D. p=q=1/2

Question No:34 **(Marks:1)** **Vu-Topper RM**

Interval estimation and confidence interval are:

A. Same
B. Opposite
C. Different
D. Independent

Question No:35 **(Marks:1)** **Vu-Topper RM**

When N becomes indefinitely large (N tends to infinity) hyper geometric distribution follows:

Normal

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